

Report team 17

1. 10 Points about the thesis.

1. The author states his motivation for this work, object of the investigation (quadrotor Linkquad) and the goal, which is to design an autonomous control loop for a given quadrotor.
2. To achieve the stated goal the author must model the quadrotor. This includes the derivation of adequate mathematical equations that describe the object.
3. The author proceeds with a description of existing control techniques like PID, LQR, h-infinity and nonlinear control.
4. To correctly describe a quadrotor with a model it is necessary to approximate physical parameters like moments of inertia, motor dynamic, aerodynamics properties, etc.
5. One of the crucial steps toward the control loop is a nonlinear mathematical model that describes the quadrotor that will be used to evaluate effectiveness and develop control loop .
6. To have a working control loop the author evaluated limitations of the firmware (only PI control loops). It is a crucial bridging point between conceptual MATLAB/Simulink models and actual hardware of the quadrotor.
7. As next step they described a control strategy (multiple inner and outer loops for various state variables) that will allow him to implement complete position control. From the chosen strategy he designed a mixed strategy for sending commands to engines.
8. The final step of developing a control loop is linearization of the mathematical model, derivation of the state equations and testing control loop with vertical, yawing, and longitudinal movement.
9. Testing of the designed control loop was done with the attitude controllers due to various limitations of available software and hardware at the time of initial development.
10. The author concludes by stating that his work was successful but too simplified due to various reasons. Despite simplicity, the autonomous control loop was successfully implemented and tested on real-world hardware.

2. Author's algorithm

- a) Define technical goals and requirements for control loop of quadrotor.
- b) Derive the symbolic mathematical model of the quadrotor.
- c) Analyze existing control techniques.
- d) Calculate or approximate physical parameters of the quadrotor.
- e) Assess technical parameters of hardware and firmware.
- f) Linearize mathematical model to derive state equations.
- g) Design control loop that corresponds to goal and respects limitations.
- h) Test control loop in software for various modes.
- i) Test control loop with hardware.
- j) Evaluate results.

3. Summary

The student wanted to design a control loop aka. “controller” for the Linkquad quadrotor. They used well established and understood mathematical models for aerial vehicles that exhibit 6DOF (Euler's angles, linearization, state space); they clearly understood limitations of the existing hardware and firmware and from there were able to choose the correct strategy on how to design control loops. As a result they developed simplified but working control loops (PI control loops) for a given quadrotor.

4. Nonlinear model

In attachment *.m files of nonlinear models from Chapter 5 of the aforementioned thesis. But for the ease of review they will be replicated here as Figures.

```
% Define the symbolic variables
mass_quadrotor = 1.07;
g = 9.81;
lever = 0.214; % meters

% The moment of inertia for the whole body
I = [ 0.0093, 0, 0;
      0, 0.0092, 0;
      0, 0, 0.0151
    ];

% thrust coefficient
b_thrust_coeff = 1.5108*10^(-5);

% inertia of the whole propeller is
Ip_inertia = 4.439*10^(-5);

% moment of inertia
Jl_mom_inertia = 4.927*10^(-6);
Jm_mom_inertia = 2.506*10^(-6);
%time constant
taunp = 0.5;
% drag factor
d_grag_factor = 4.406*10^(-7);
```

Figure 1. Symbolic values used in modeling.

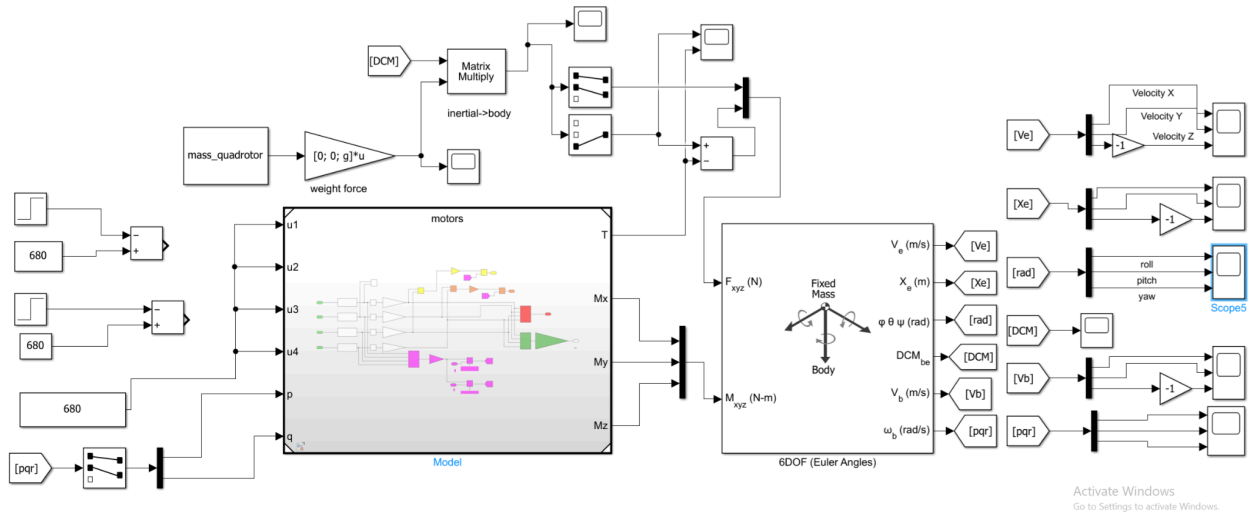


Figure 2. Simulink model of the quadrotor.

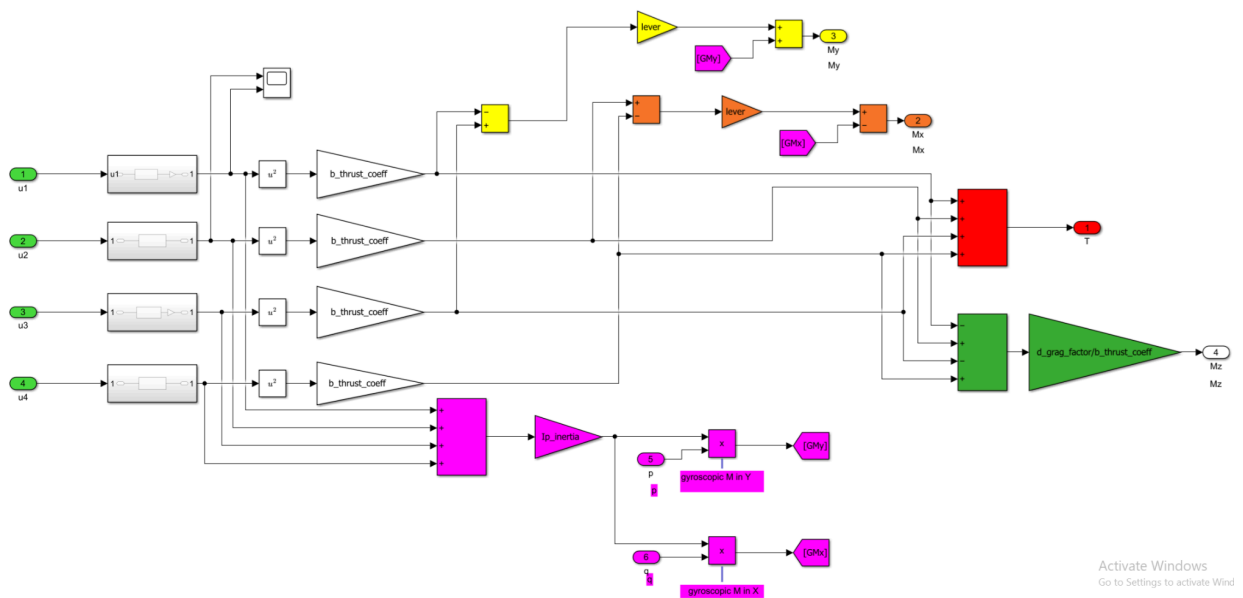


Figure 3. Simulink model of the motor subsystem.

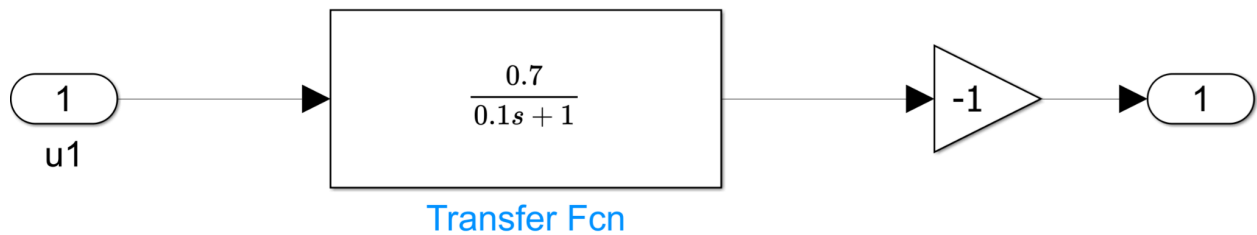


Figure 4. Simulink model of the motors (1 - 3). Gain module was added to simulate opposite direction of rotation to motors 2 and 4.

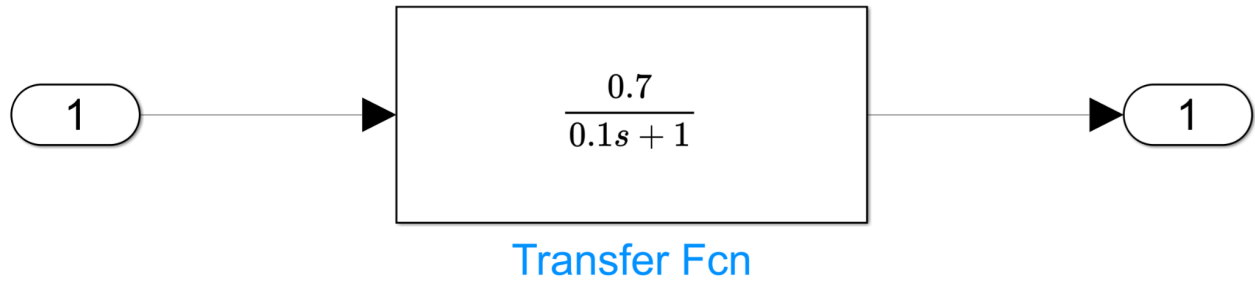


Figure 5. Simulink model of the motors (2 - 4).

4.1 Simplest input. All motors with the same value.



Figure 6. Velocity in Earth frame of reference - output from 6DOF subsystem. Motors values are 680.

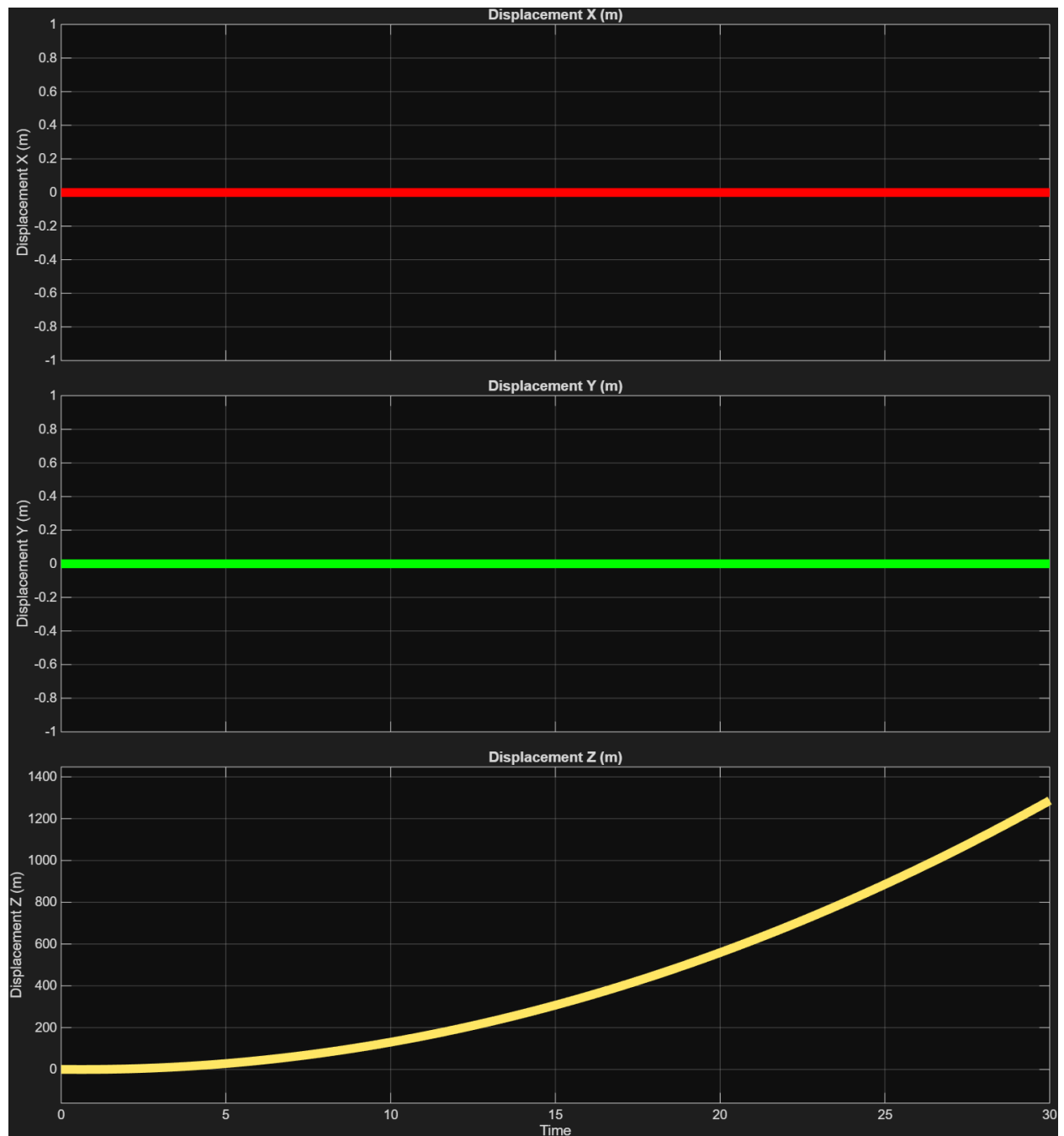


Figure 7. Position along X,Y,Z coordinates in Earth frame of reference - output from 6DOF subsystem. Motors values are 680.

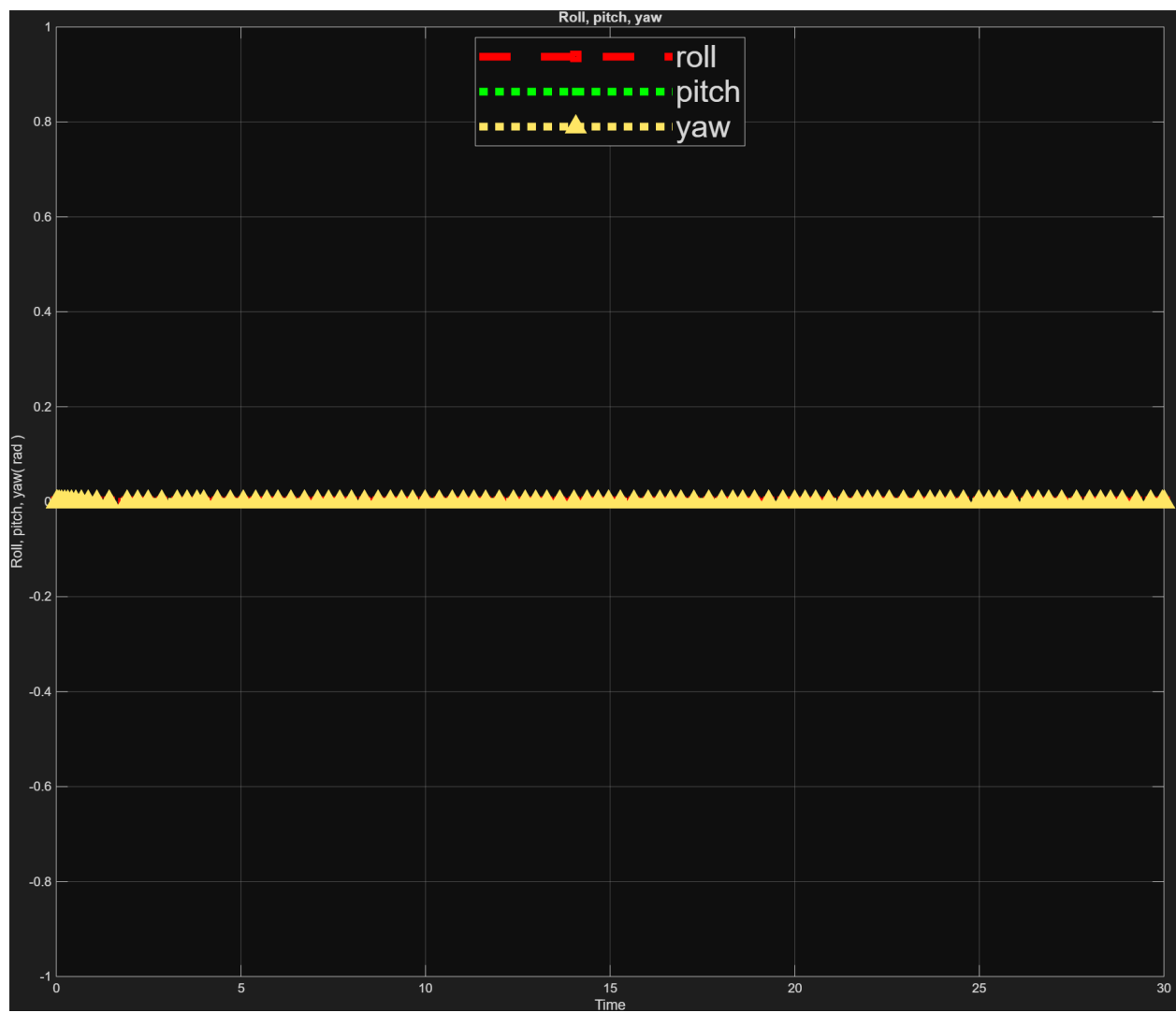


Figure 8. Euler angles - output from 6DOF subsystem. Motors values are 680.

Оформіть будь ласка результати у формі звіту.

Для кращої репрезентативності:

1. можна створити табличку зі сценаріями по 6 степенях свободи,
2. додати очікуваний рух від комбінації команд,
3. додати ізольовані графіки які показують зміни кутів та координат,
4. Усі блок схеми та скріншоти моделі додатки
5. Більш репрезентативні графіки з кращим масштабом

4.2 UAV is 6-DOF an object.

Quadrotor is 6-DOF an object that is controlled by 4 rotors placed on its principal axis. Rotors 1-3 are on the X axis and rotors 2-4 on the Y axis. Therefore the following possibilities are possible to move it in all 6 directions.

Let's denote relative speeds of rotors with following symbols:

- + $\rightarrow$ rotates faster
- $\rightarrow$ rotates slower
- = $\rightarrow$ equal speed

And a position in sequence denoting rotors number, for example **===-** would mean that rotors 1-3 have the same speed and rotor number 4 is slower than the rest.

Equations that govern momentums around the X,Y, Z axis are similar to one presented in the master thesis except for M_y .

To adhere to right hand rule momentum should be:

$M_y \propto T_3 - T_1$ and **not $T_1 - T_3$** (where T_1 thrust generated by motor 1; T_3 by motor 3)

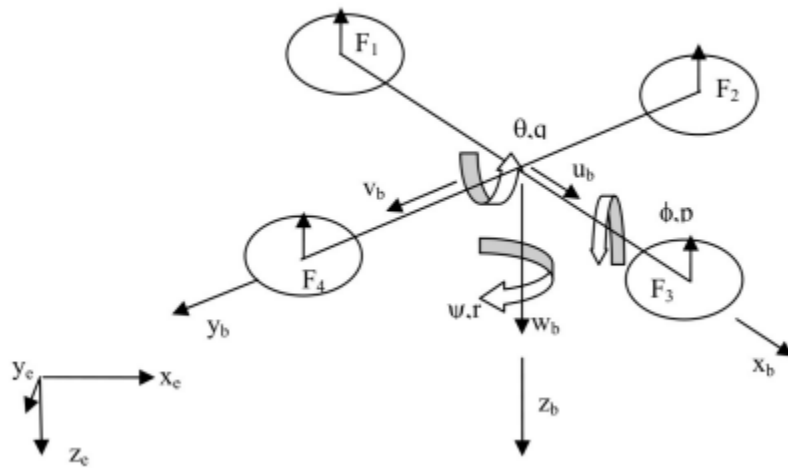


Figure 9. Rotors and principal axis.

Table 1. Commands and expected direction of movement in 6-DOF..

Rotors speed and command	Expected direction of movement	Explanation
====	+Z	With sufficient speed thrust generated by rotors will overcome gravity.
====, 0000	-Z	Quadrotor will free fall from reached height.
+=== or ==- =	+X	Difference creates negative momentum around the Y axis. ($T_3 < T_1$)
==+= or -===	-X	Difference creates positive momentum around the Y axis. ($T_3 > T_1$)
=+= or ==- =	+Y	Difference creates positive momentum around the X axis. ($T_2 > T_4$)
===+ or -= =	-Y	Difference creates negative momentum around the X axis. ($T_2 < T_4$)

Direction +X.

Speed set on rotors: 690 680 680 680 (+===)

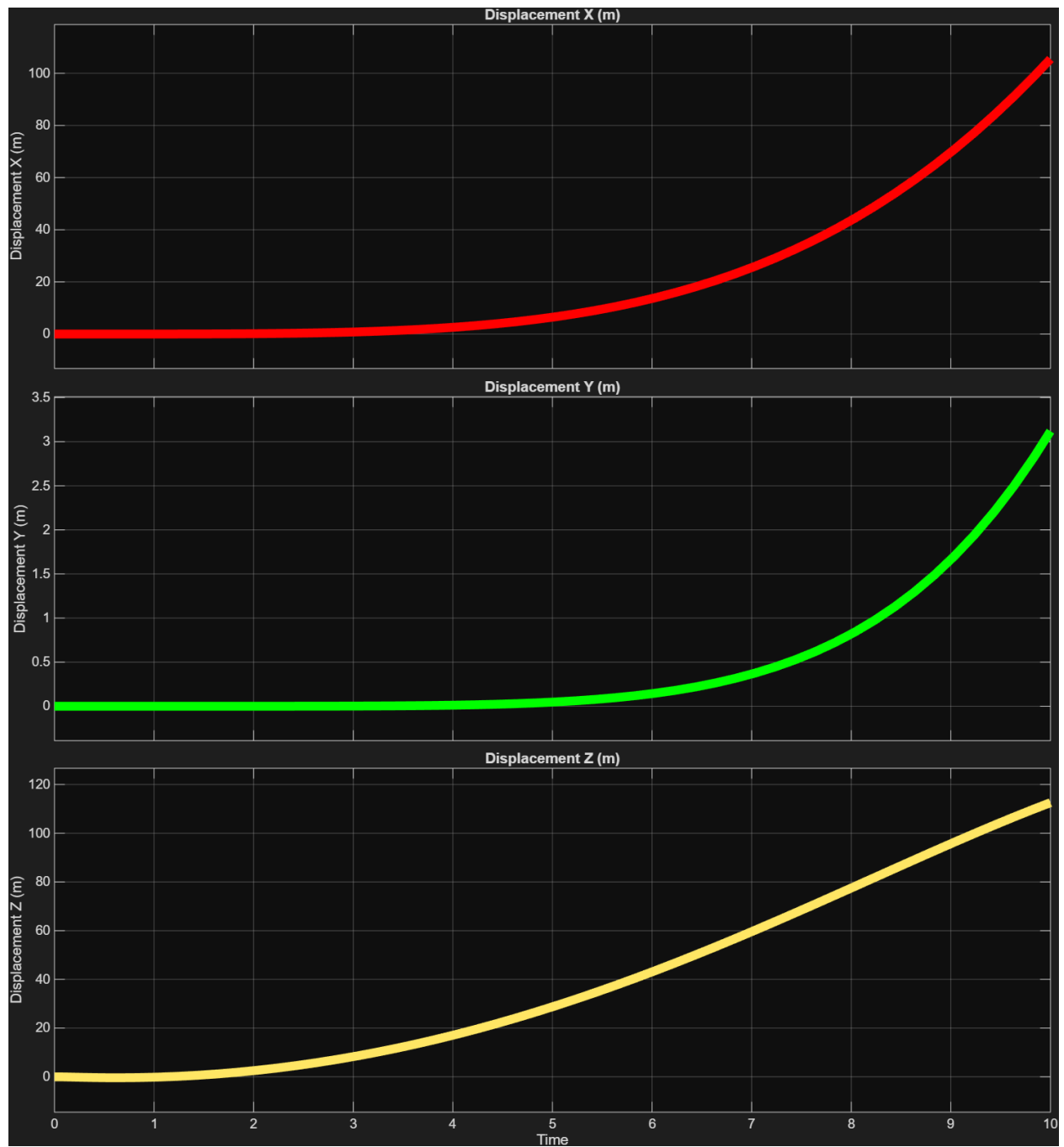


Figure 10. Isolated movement along +X direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect Y also drifts.

Direction +X

Speed set on rotors: 680 680 **670** 680 (==-=)

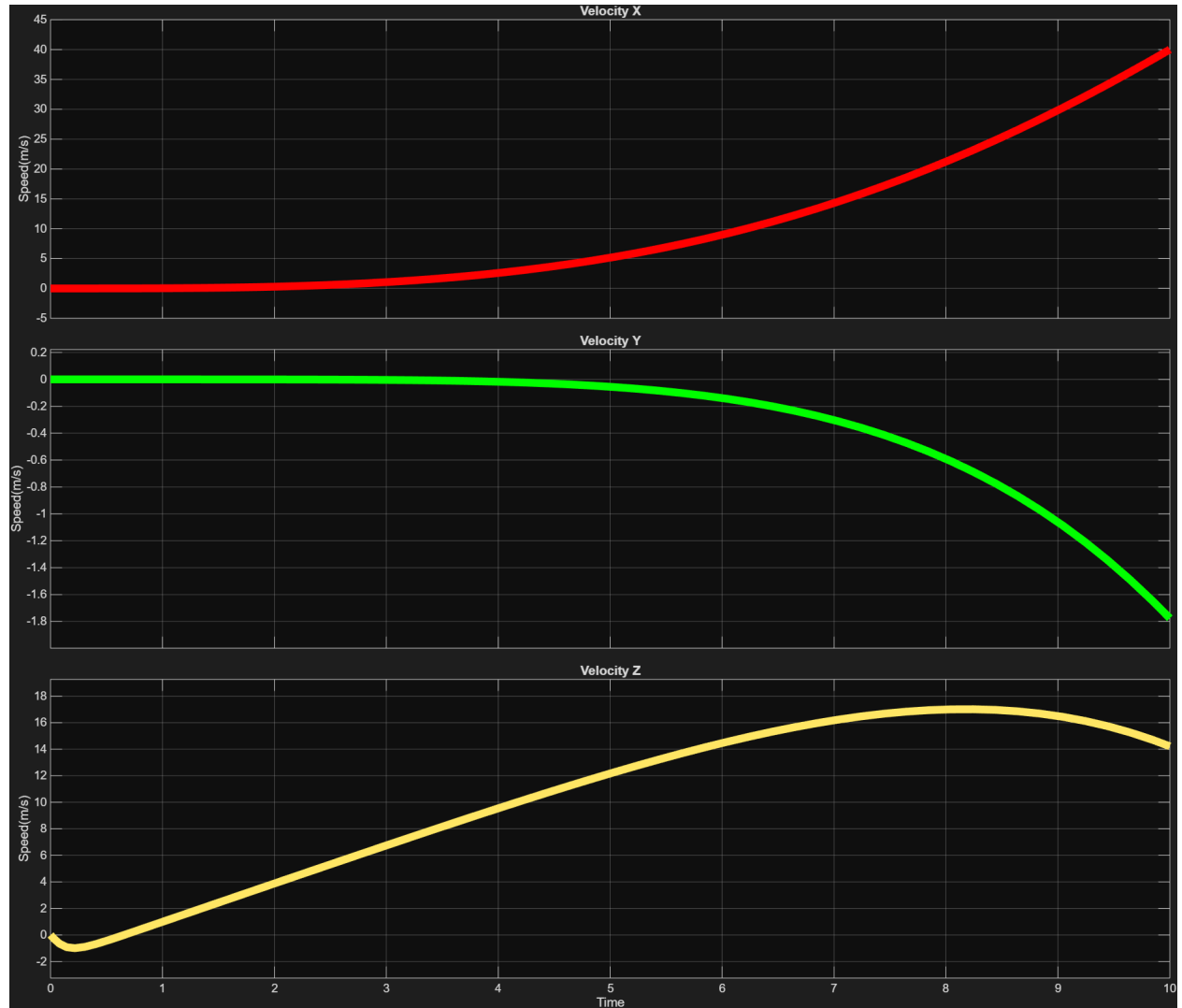


Figure 11. Isolated movement along +X direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect Y also drifts.

Direction -X

Speed set on rotors: 680 680 **690** 680 (==+=)

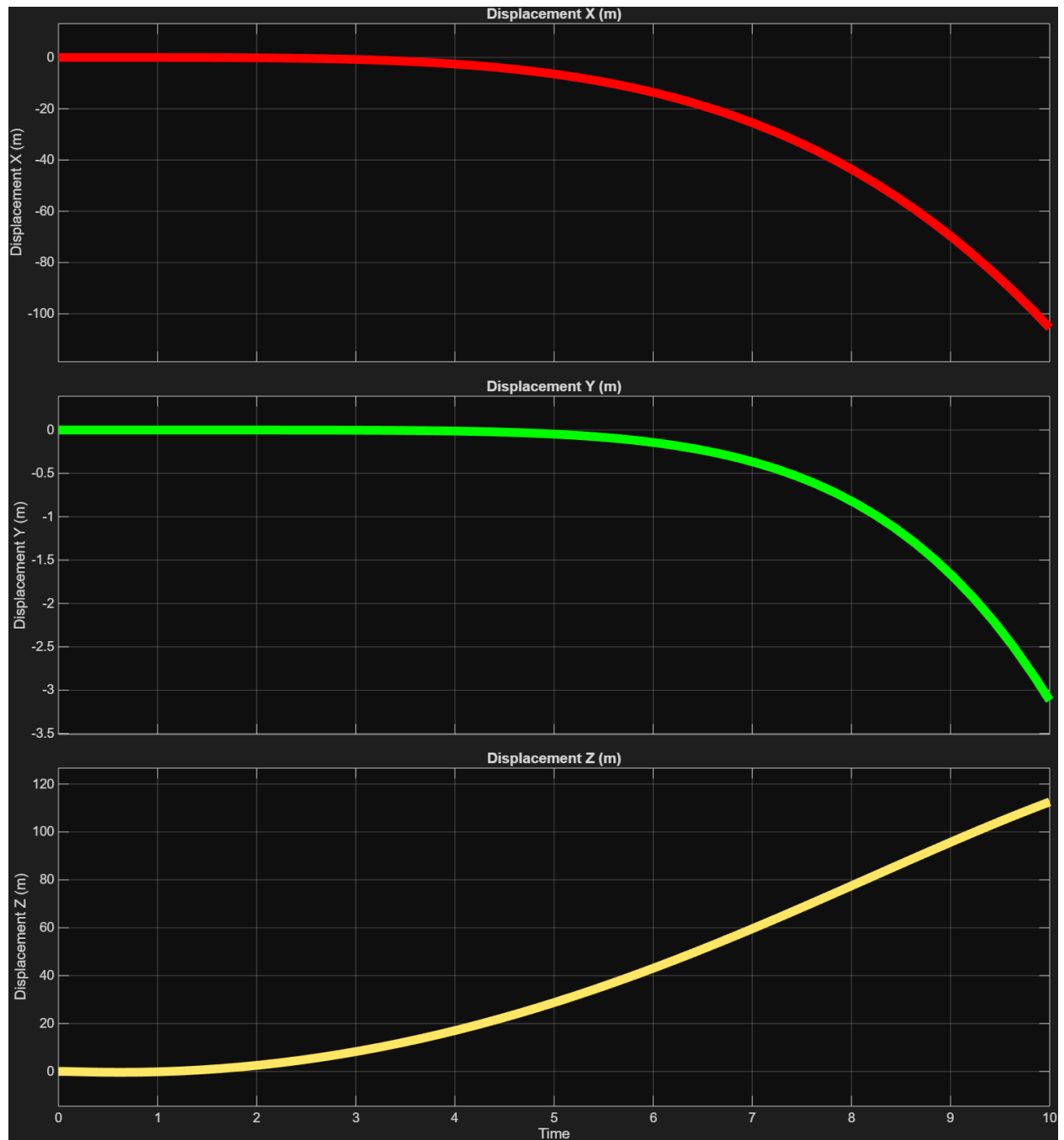


Figure 12. Isolated movement along -X direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect Y also drifts.

Direction -X

Speed set on rotors: 670 680 680 680 (==)

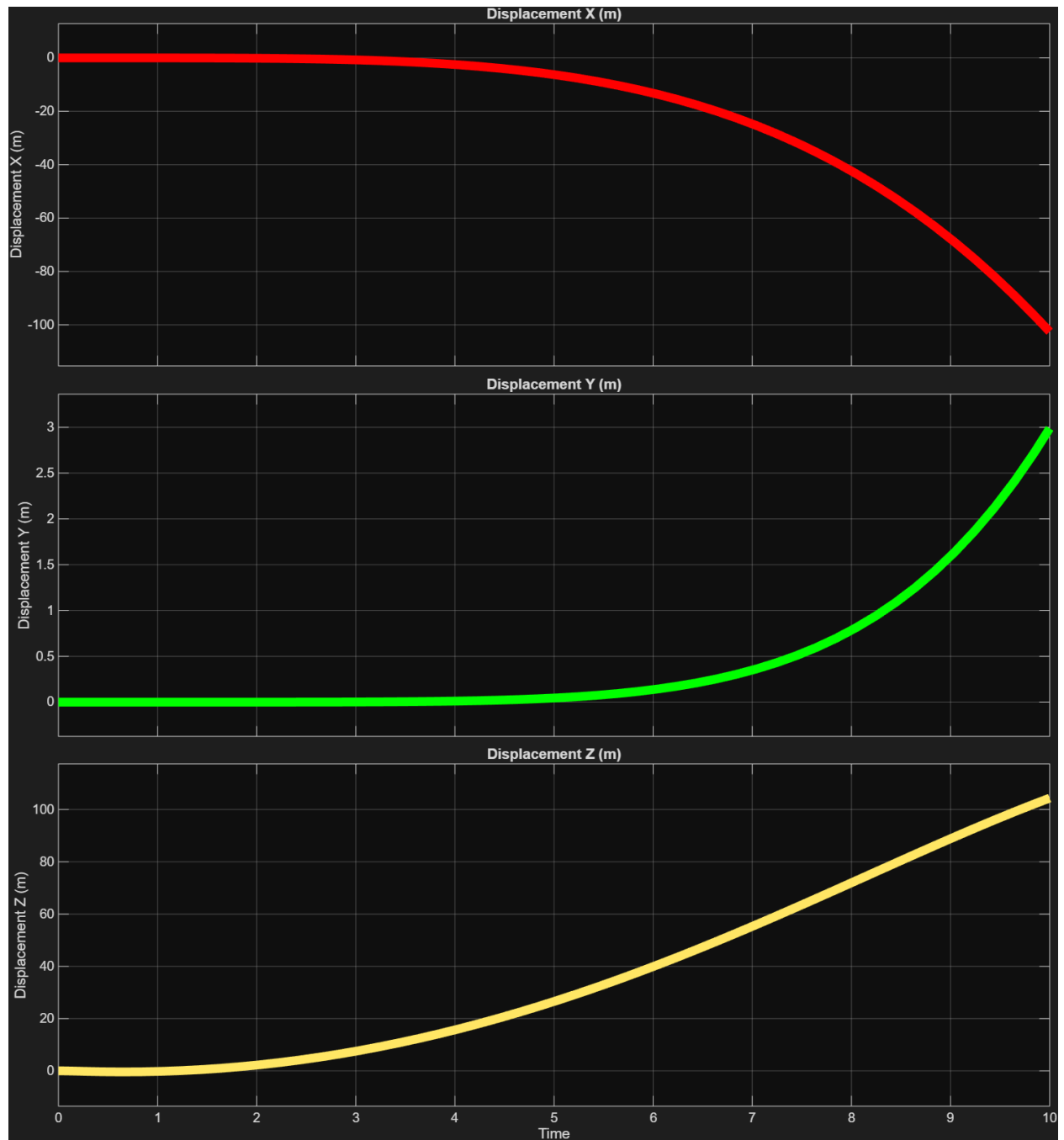


Figure 13. Isolated movement along -X direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect Y also drifts.

Direction +Y

Speed set on rotors: 680 **690** 680 680 (=+==)

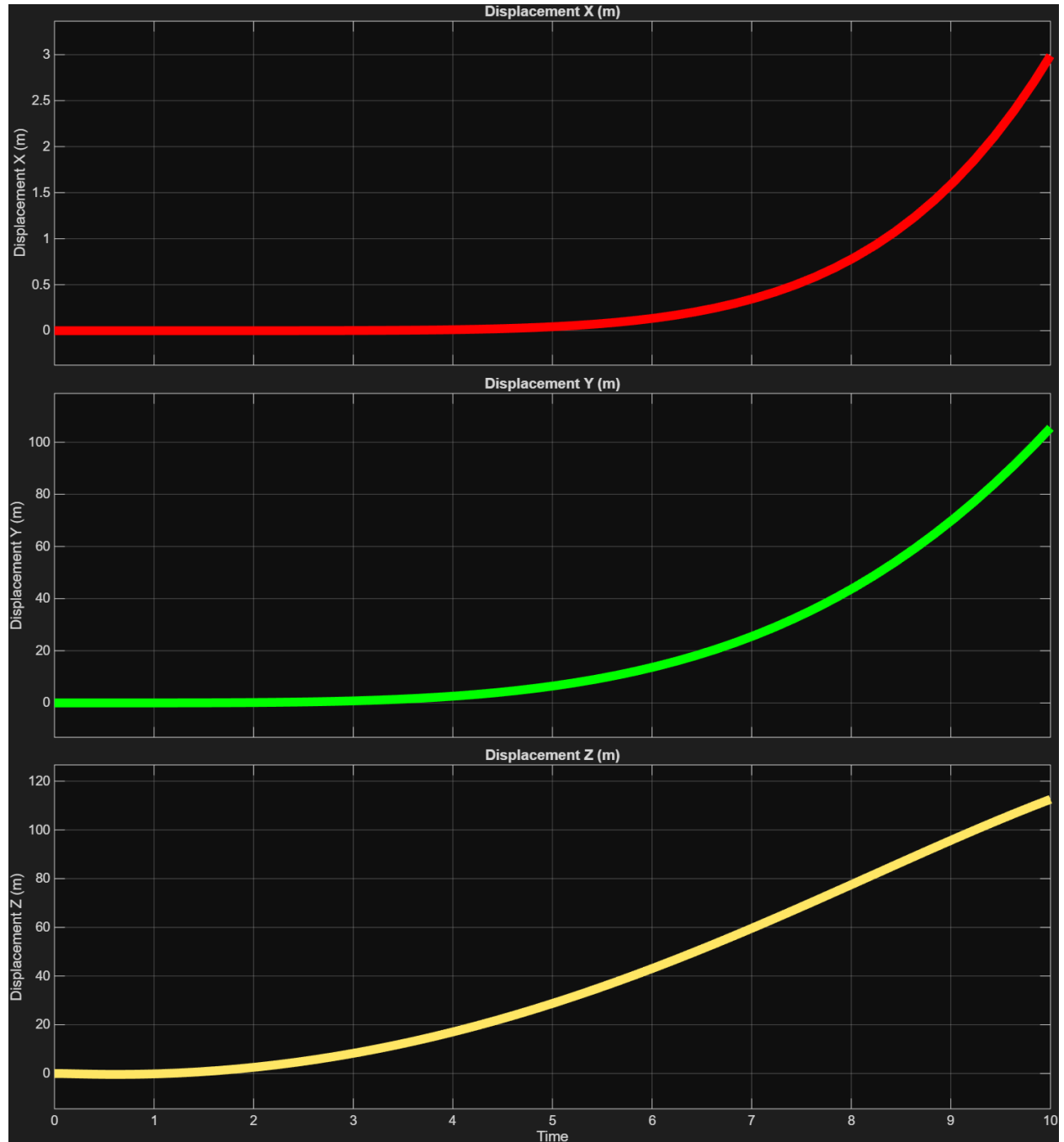


Figure 14. Isolated movement along +Y direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect X also drifts.

Direction: +Y

Speed set on rotors: 680 680 680 **670** (==--)

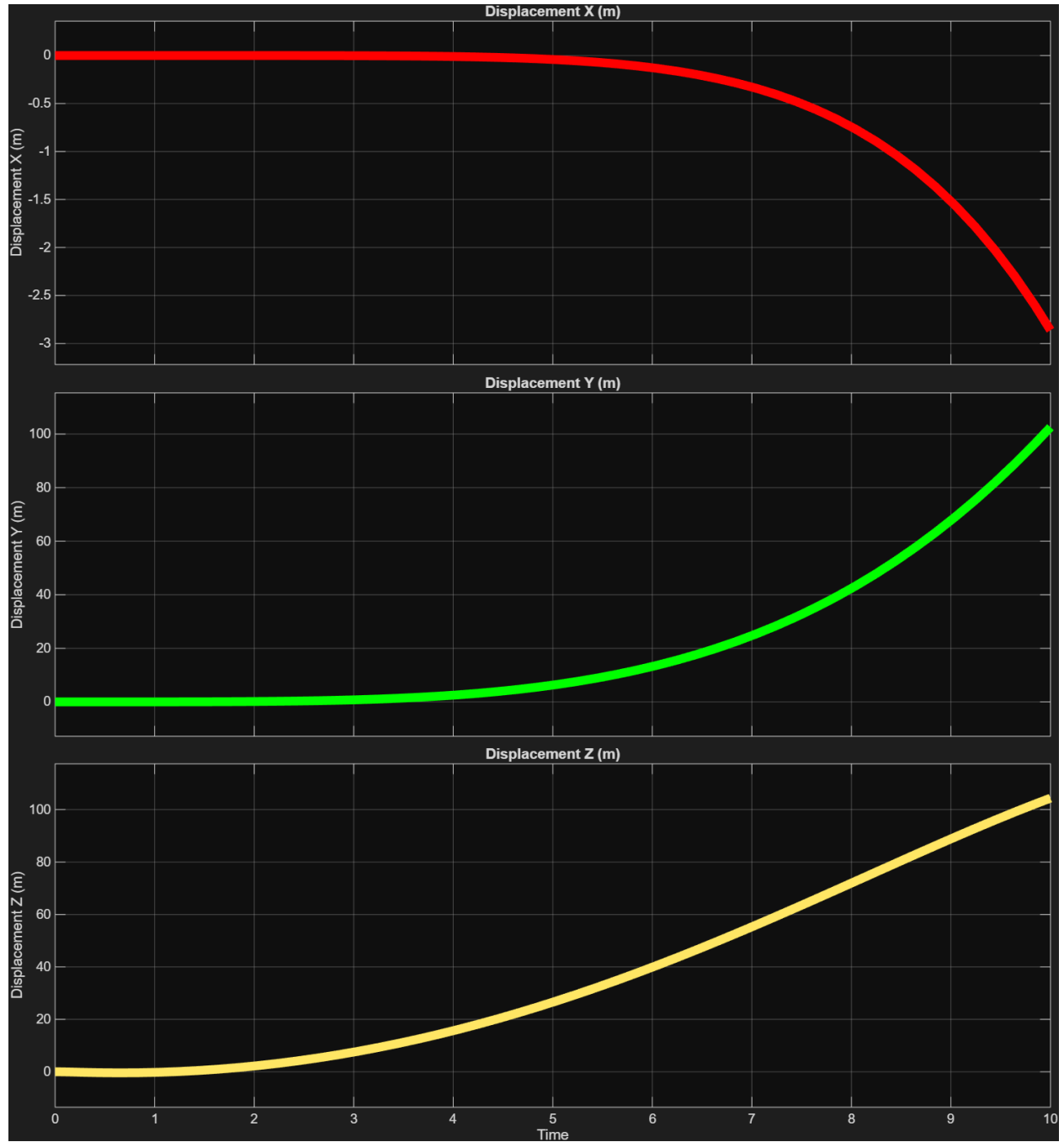


Figure 15. Isolated movement along +Y direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect X also drifts.

Direction: -Y

Speed set on rotors: 680 680 680 **690** (===+)

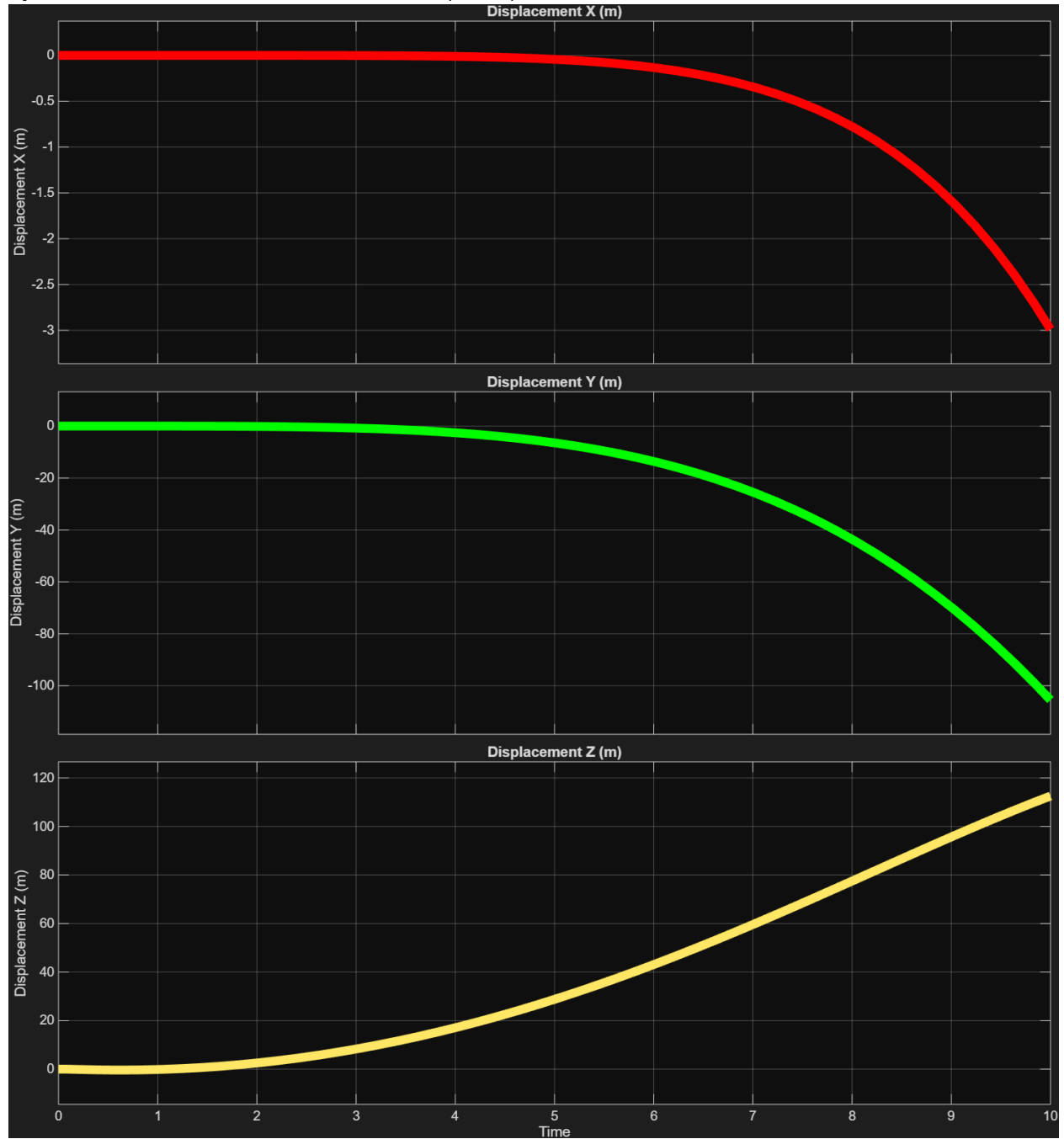


Figure 16. Isolated movement along -Y direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect X also drifts.

Direction: -Y

Speed set on rotors: 680 670 680 680 (==)

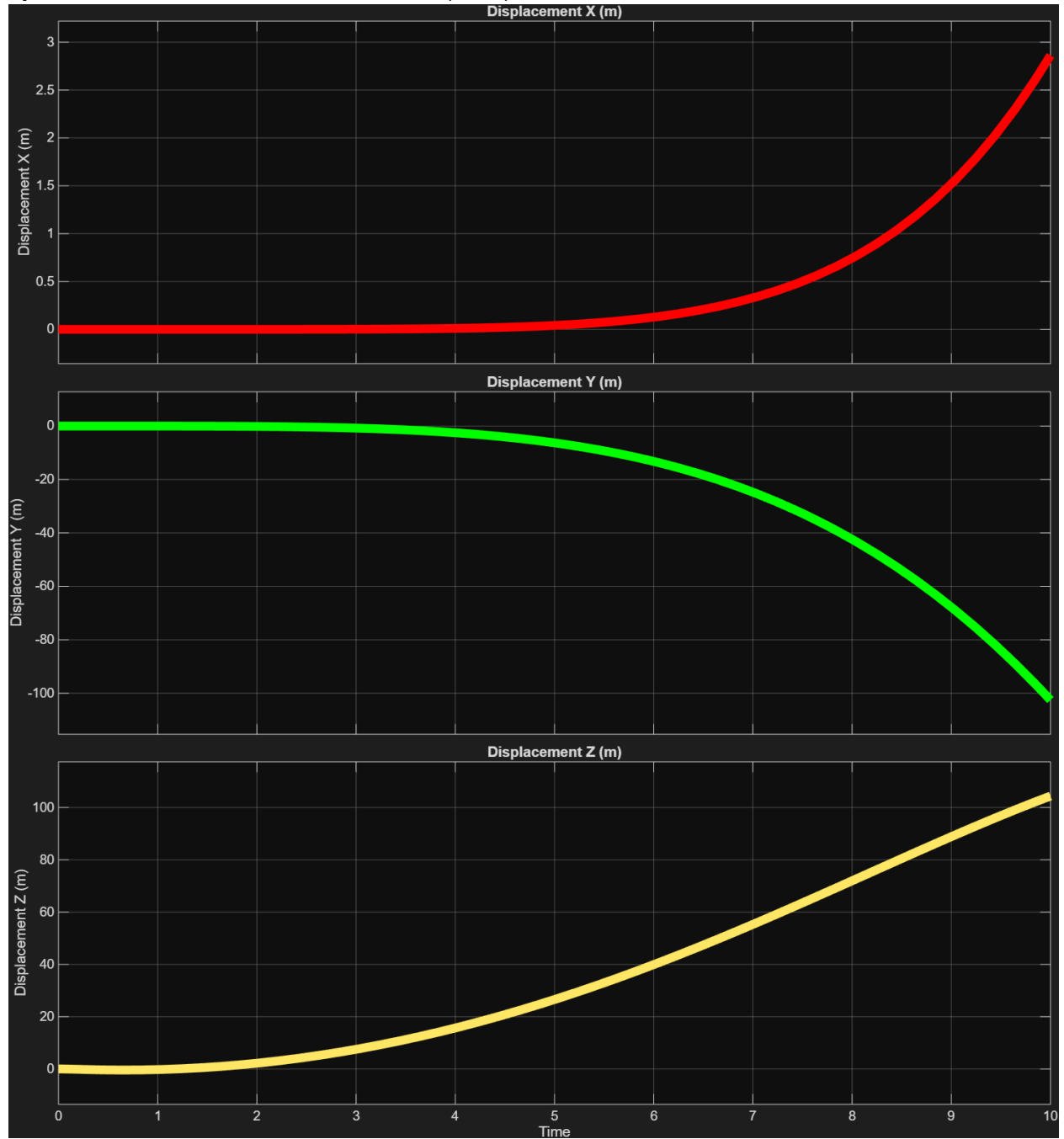


Figure 17. Isolated movement along -Y direction. Because of positive thrust, the Z coordinate also grows and due to coupling effect X also drifts.

4.2 Euler Angles

4.2.1 Isolated roll

We would like to simulate roll change in a quadrotor. This is achieved by introducing imbalance between rotors speed 2 and 4. To have better isolated change in roll we would need to keep thrust almost the same and to achieve that we introduce step function (step time 4 seconds ; step value 10), which will simultaneously increase speed on one rotor but decrease to the same degree on another.

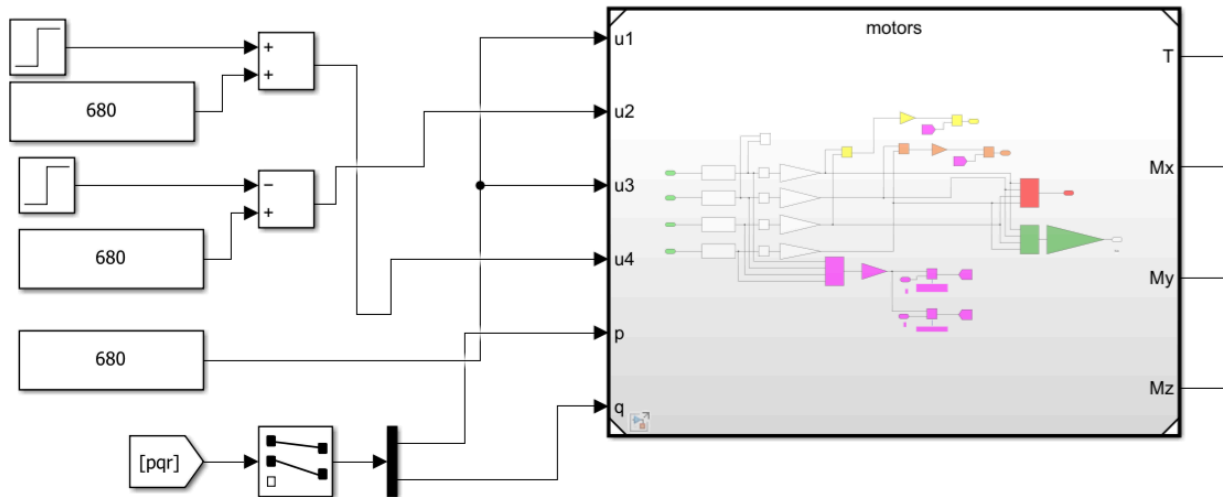


Figure 18. Isolated roll in simulink model.

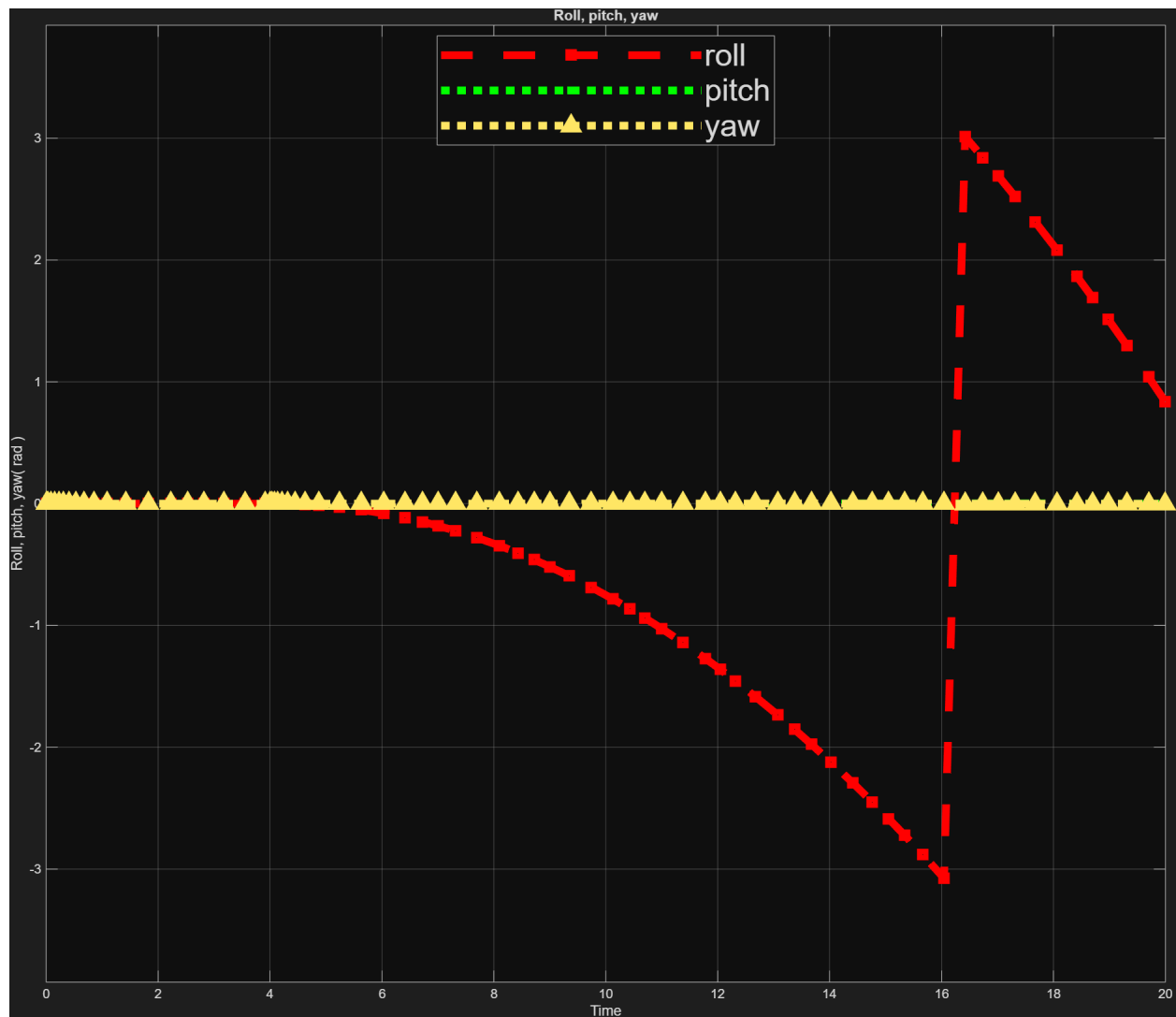


Figure 19. Isolated roll. When the bigger speed is on rotor 4.

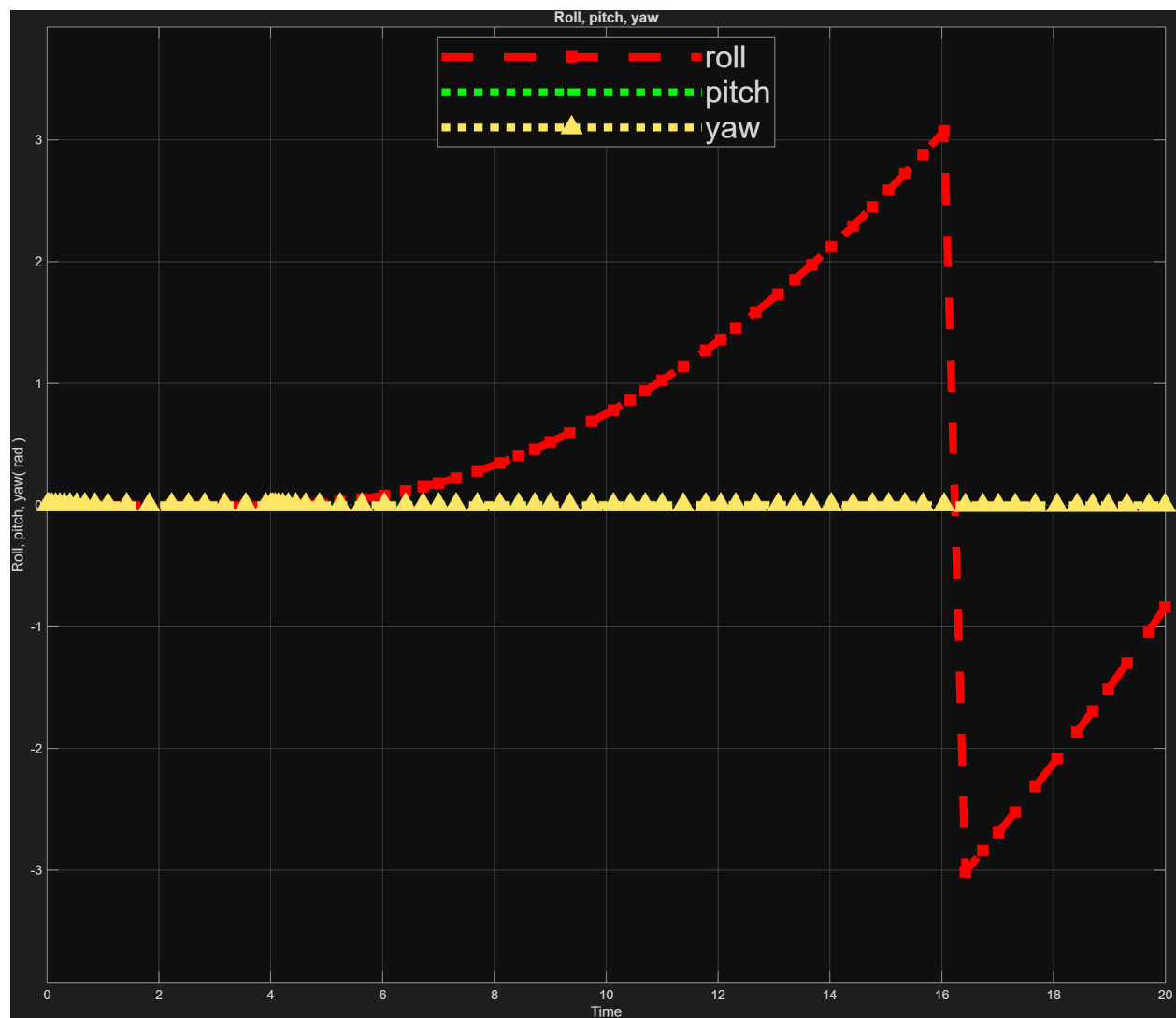


Figure 20. Isolated roll. When the bigger speed is on rotor 2.

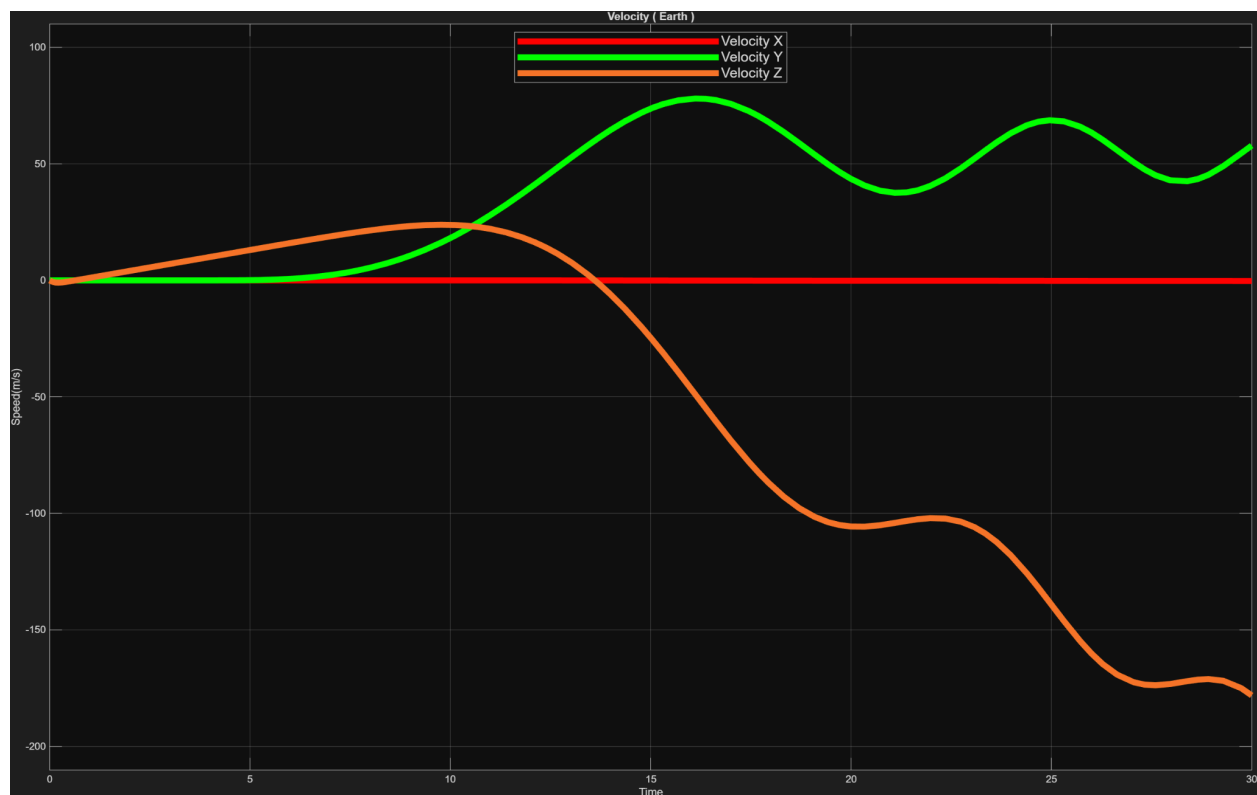


Figure 21. Velocity (Earth frame) during isolated roll.

4.2.1 Isolated pitch

Similar module as for isolated roll but motors that are changing are 1 and 3 (Step time 4 seconds ; step value 10).

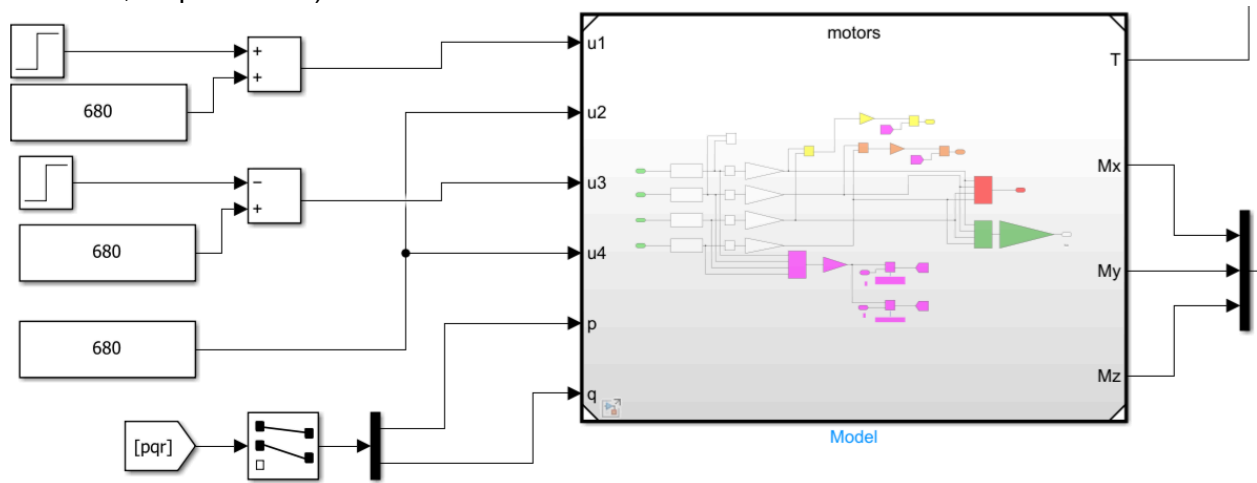


Figure 22. Isolated roll in simulink model.

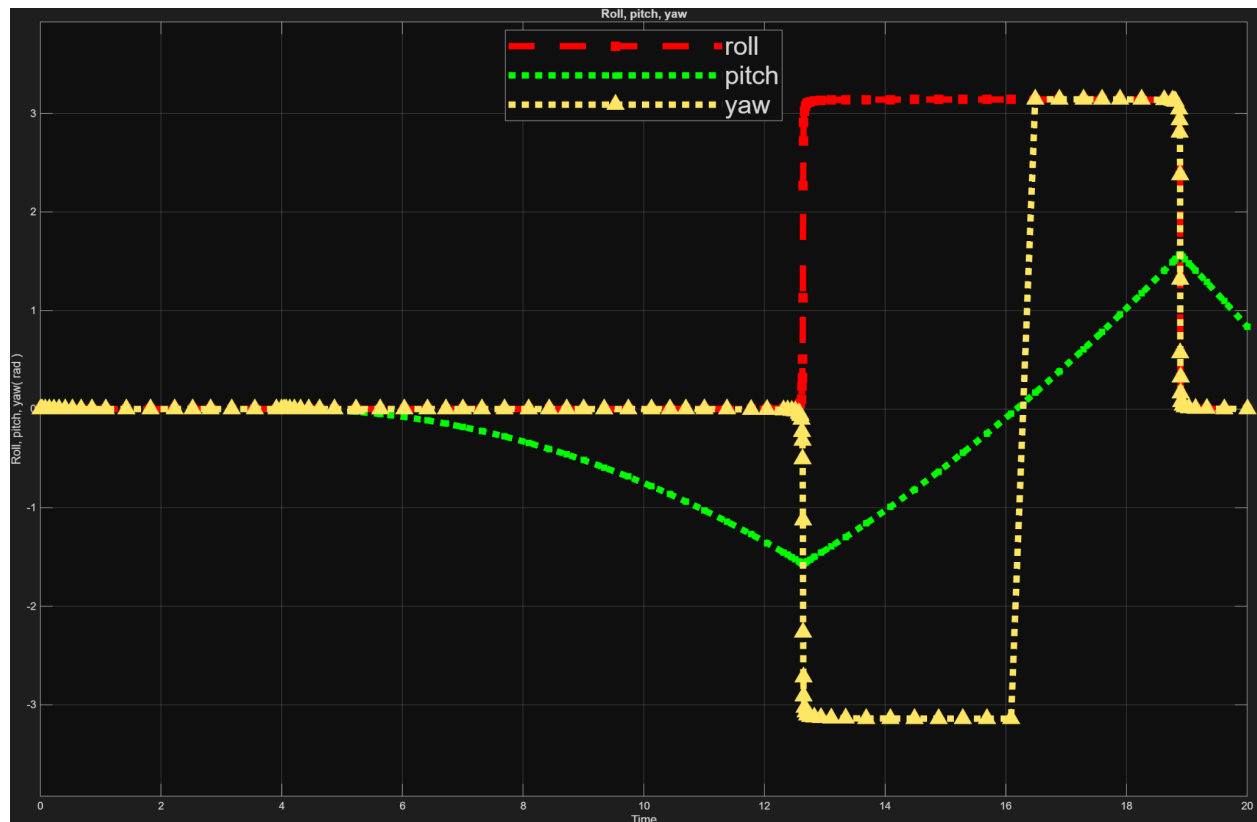


Figure 23. Isolated pitch. When the bigger speed is on rotor 1. Roll and Yaw flips due to **gimbal lock** effect which happen in Euler angel system with pitch ± 90 degrees which causes X axis to be parallel to Z and that in turn makes subsequent Roll or Yaws indistinguishable from each

other (aka. lock) and in modeling it occurs as a sudden flips. To avoid this phenomenon, use of quaternions is advised.

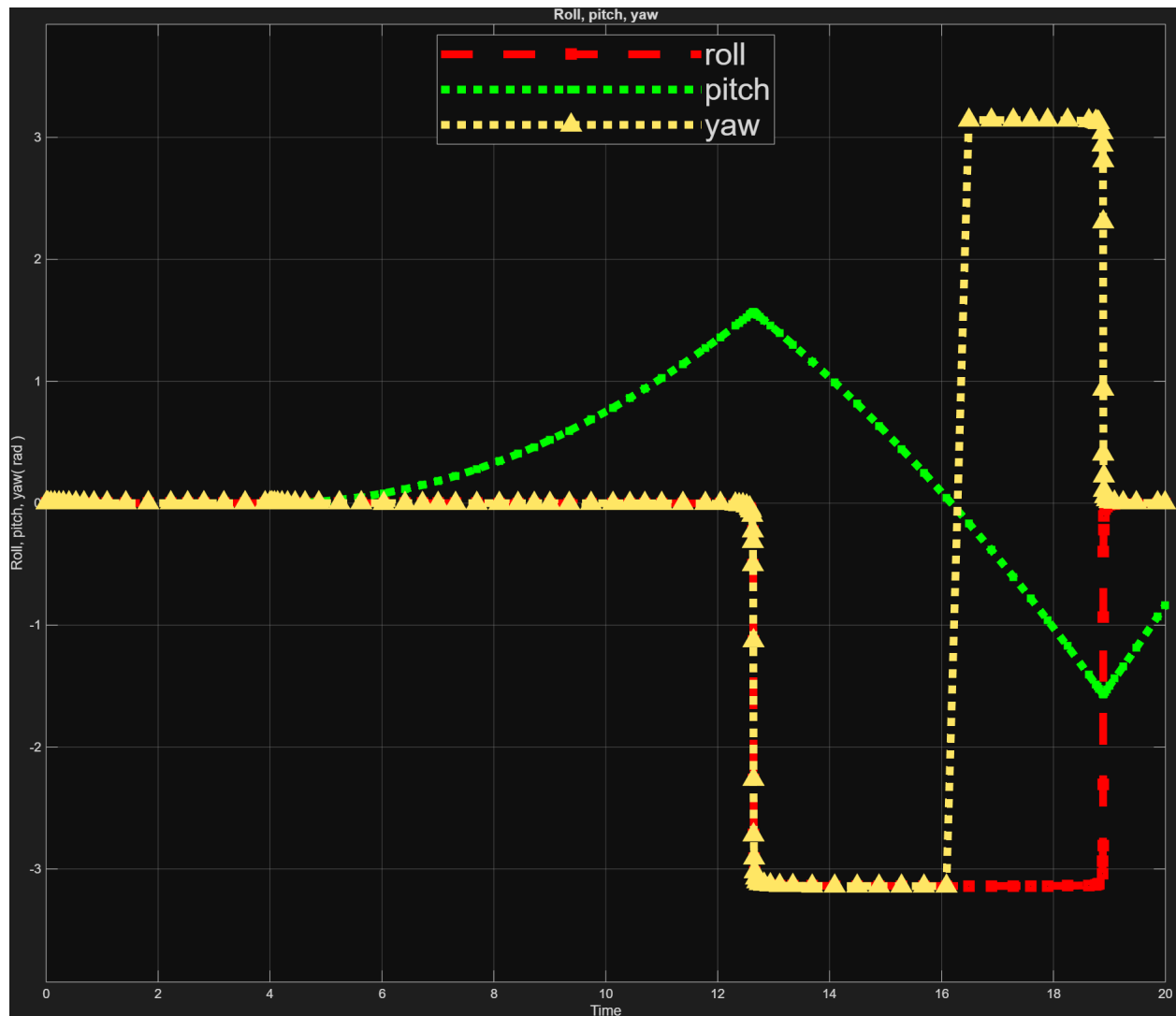


Figure 24. Isolated pitch. When the bigger speed is on rotor 3. With gimbal lock in different direction as well.

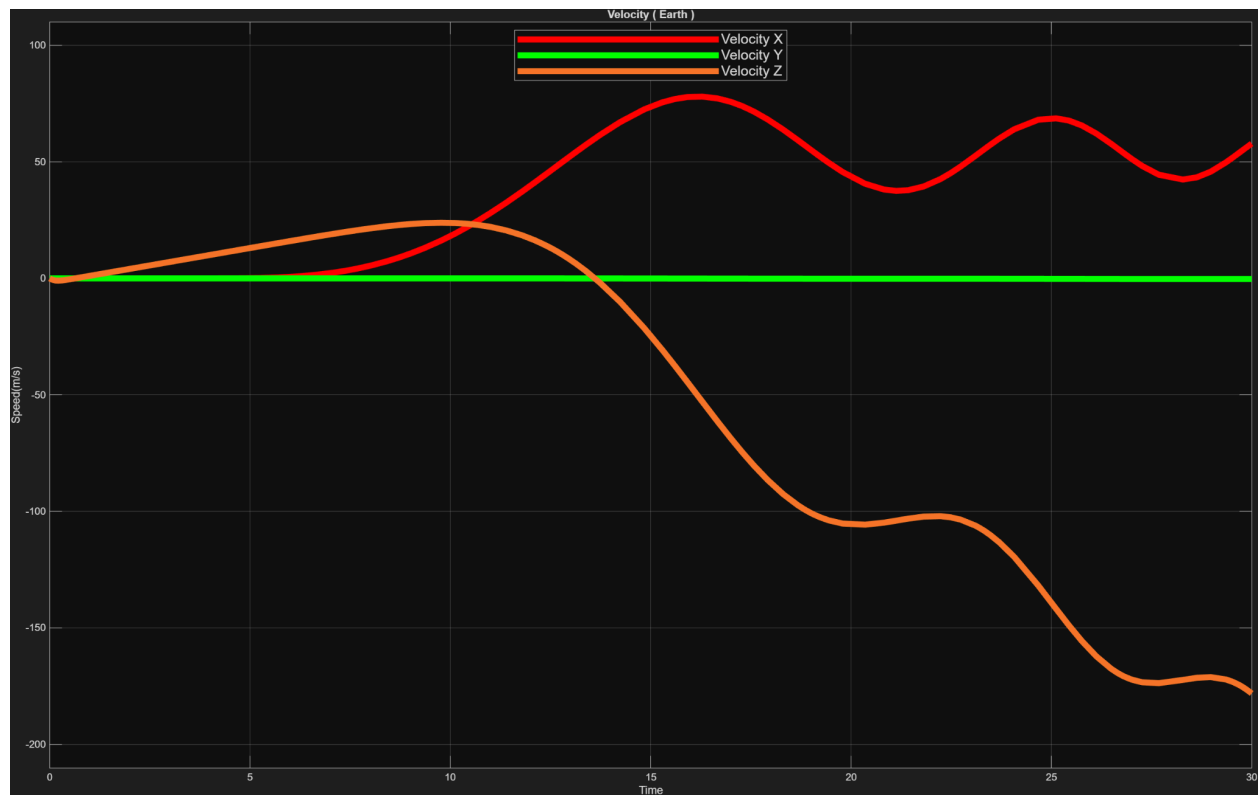


Figure 25. Velocity (Earth frame) during isolated pitch.

4.2.3 Isolated yaw

Yaw is achieved by 3rd Newton's law. In our system this means that either clock wise or counter clock wise rotors are spinning faster (or slower) this imbalance creates momentum in the opposite direction to the faster spinning pair. To have yaw values grow faster step value was 30 and step time 4.

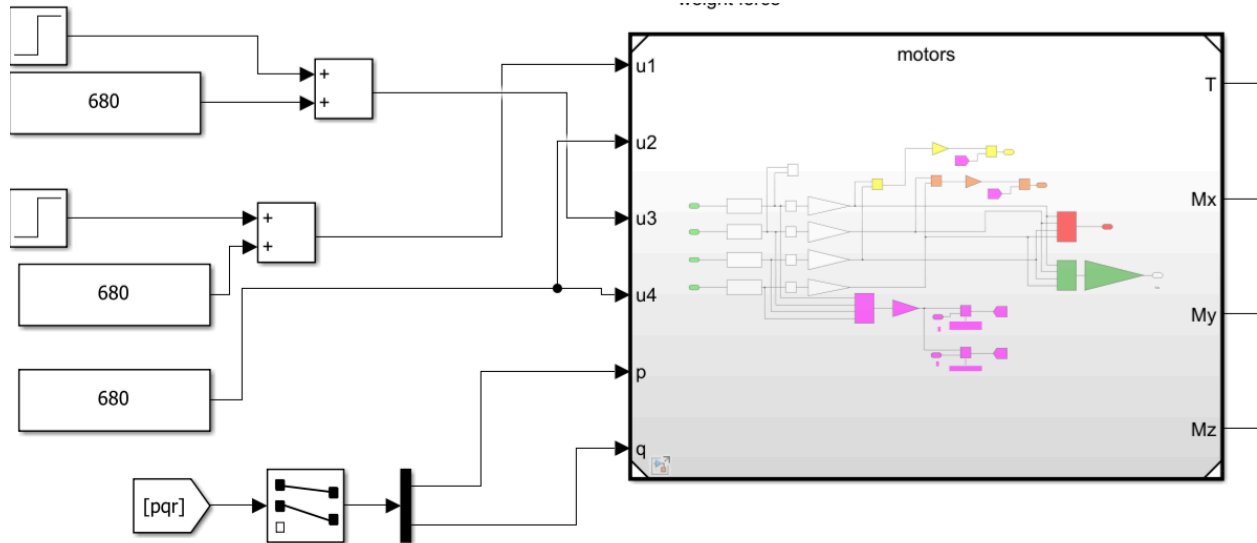


Figure 26. Isolated yaw in simulink model. We use rotors 1 and 3 to control yaw.

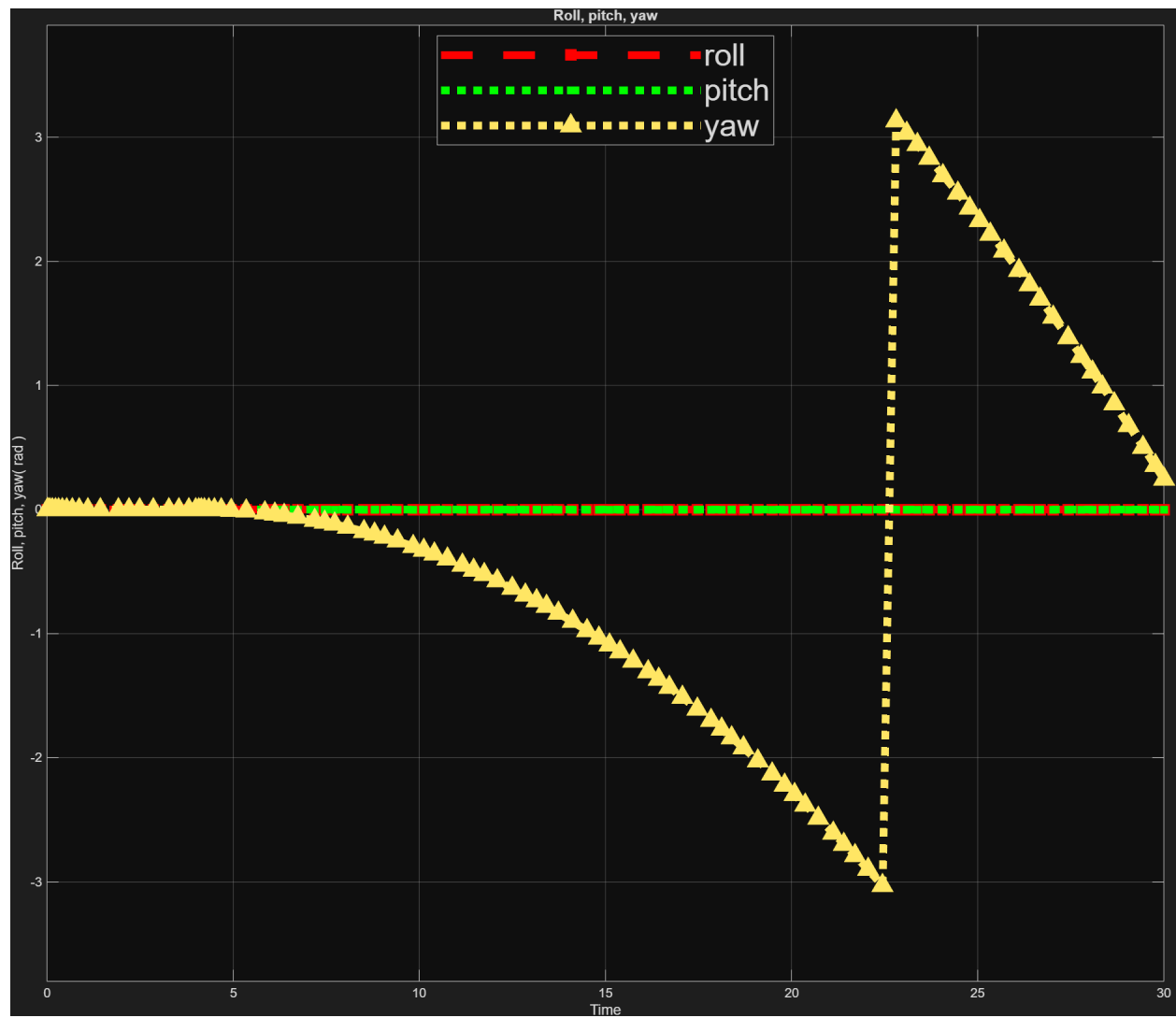


Figure 27. Isolated yaw. When the bigger speed is on rotors 1 and 3.

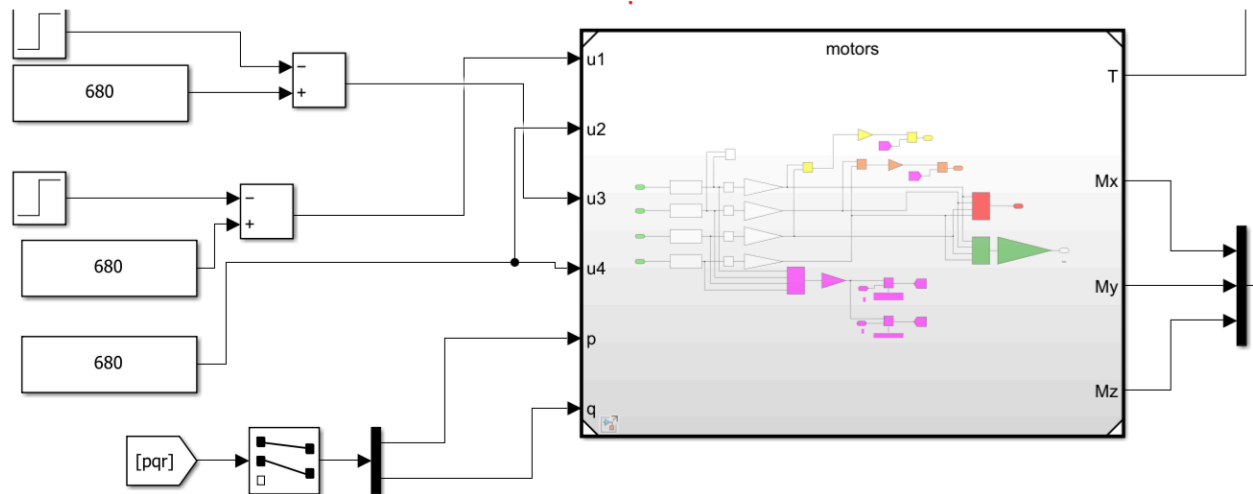


Figure 28. Isolated yaw in simulink model. We use rotors 1 and 3 to control yaw but by reducing speed on them.

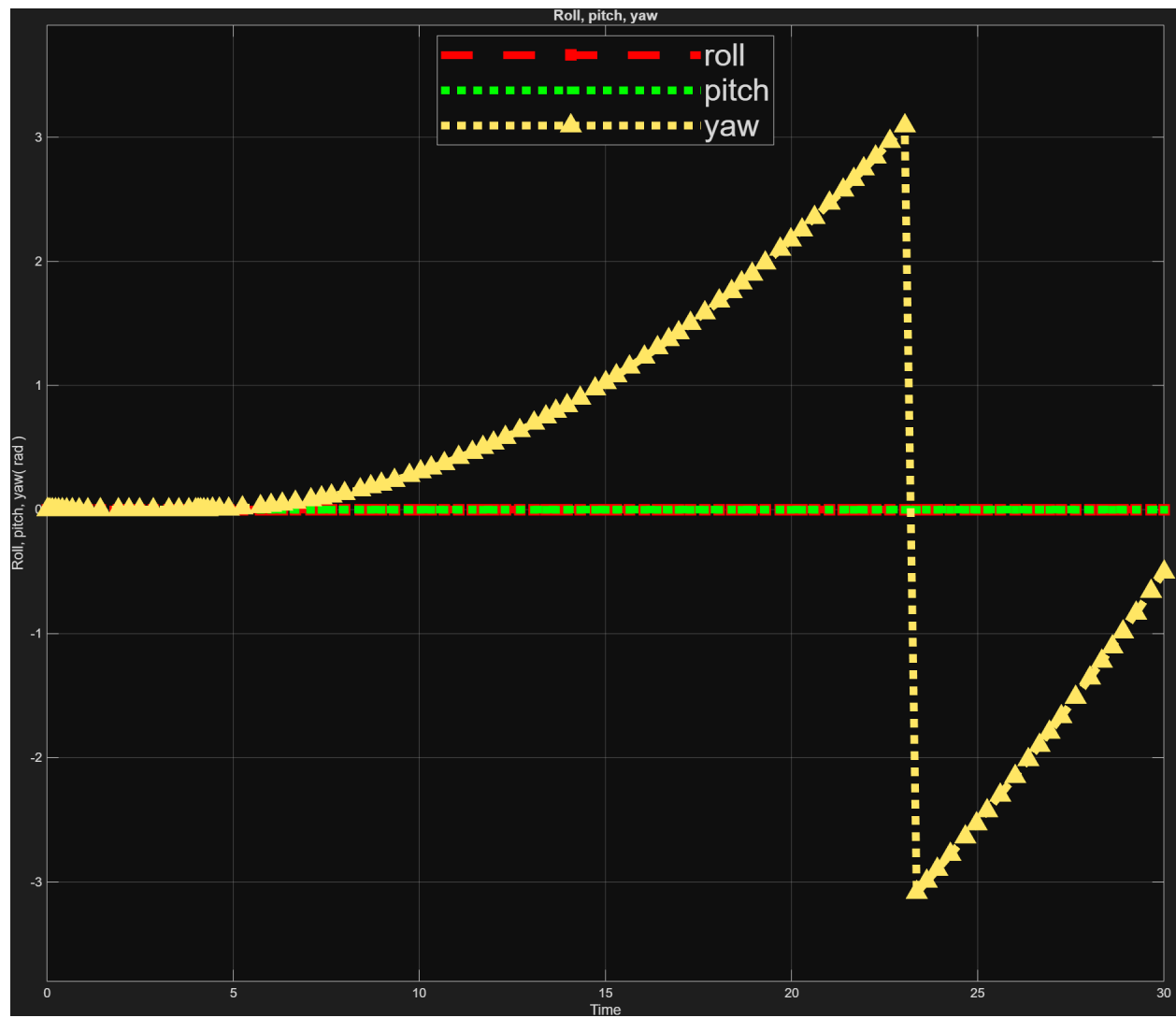


Figure 29. Isolated yaw. When the bigger speed is on rotors 2 and 4 because speed on 1 and 3 was reduced.



Figure 30. Velocity (Earth frame) during isolated yaw.

5. Conclusion

The thesis provides an accurate model of the quadrotor. All inputs cause the expected maneuvers and mathematical limitations, such as gimbal lock. The model can be used for further development of the control loop. The only necessary modification is presented in Figure 3 and explained in Section 4.2, where reversing the signs for rotors 3 and 1 was required to calculate moments that align with the right-hand rule.